

The empirical recurrence for OEIS A251454: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A251454 records “Empirical recurrence of order 99”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 256$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 99, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 99 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A251454 is “Number of $(n+1) \times (3+1)$ 0..3 arrays with every 2×2 subblock summing to a prime.”. It has offset 1 and begins

6398, 194586, 5783416, 172965058, 5162094982, 154267520858, 4606843590302, 137657116629832, ...

The entry states:

Empirical recurrence of order 99 (see link above).

As of the “Last modified” line on the live entry (May 24 2024 13:20:24, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 256$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 256$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 512$ exact terms of the model returns order 99 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{99} c_i a(n-i),$$

c_1	10	c_{26}	21606741917843425918711910333	c_{51}	−68129329191
c_2	2143	c_{27}	−34132141411296796685503431203	c_{52}	−119550002870
c_3	−17247	c_{28}	−557183899687394998555768729449	c_{53}	154153685474
c_4	−1903602	c_{29}	790533750155893993863413578007	c_{54}	3705674894628
c_5	12713505	c_{30}	12045630964397718836450547000824	c_{55}	−201986795479
c_6	943472824	c_{31}	−15315765994420855736366555207090	c_{56}	−973398195980
c_7	−5305451728	c_{32}	−219375311763439702357127744624320	c_{57}	−14929050943
c_8	−296861717465	c_{33}	249044317260277752335154249811701	c_{58}	2159466916931
c_9	1419685529732	c_{34}	3378896183405310469835126947812612	c_{59}	1738628119156
c_{10}	63984074164294	c_{35}	−3406616747941649528357215181668311	c_{60}	−4029367642656
c_{11}	−262748621185742	c_{36}	−44151786200451431048637913373121234	c_{61}	−568062604469
c_{12}	−9942923459777715	c_{37}	39247355752003944542641799986189444	c_{62}	6292248646228
c_{13}	35404642133665835	c_{38}	490642607314317001708088189402931867	c_{63}	1238138849935
c_{14}	1156021782844313512	c_{39}	−380881125877799217040253208831468505	c_{64}	−8175277723867
c_{15}	−3602775896598226604	c_{40}	−4645305210363466486157021198812210156	c_{65}	−2027532820443
c_{16}	−103412259359368424306	c_{41}	3110009703844130722853012518230681453	c_{66}	8776736320717
c_{17}	284412908368942039475	c_{42}	37517686866468245419373319931386161472	c_{67}	2588545383840
c_{18}	7273634926443660395628	c_{43}	−21307556211402694000515528914955644505	c_{68}	−7723235851355
c_{19}	−17774186361682237465187	c_{44}	−258664193995520931184130699453408765536	c_{69}	−2610103997350
c_{20}	−409207356910450534062716	c_{45}	121880308143761465497095835639722399773	c_{70}	5518415268200
c_{21}	893118923857001971183998	c_{46}	152261804698713321942653233800898047810	c_{71}	20843360641236
c_{22}	18667978509904262710051431	c_{47}	−577084957249778891386969045612007511466	c_{72}	−3166510644161
c_{23}	−36523050759237198697953369	c_{48}	−7649723533868993029780996683345338866218	c_{73}	−1314533793574
c_{24}	−698254320183697270512008218	c_{49}	2227992436213967413146143734483137133152	c_{74}	14401396392509
c_{25}	1227151494024974990867488080	c_{50}	32772573694820728409465269467199183264734	c_{75}	6505034923957

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 99, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $99 + 4$ terms removed returns order 99 as well, so $\deg q_{\min} = 99 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $99 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 99 that this sequence satisfies — and that family turns out to have exactly one member.

References

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